

Birla Institute of Technology & Science, Pilani
First Semester 2025-2026 (Mid Semester Examination)
ECE F314: Electromagnetic Field & Microwave Engineering
ECE F331: Microwave Engineering
(Closed Book)

Part: A

Duration: 45 minutes

Total Marks: 30

Note to Students:

Please write the answers in the answer sheet provided with proper units and signs. Marks will be deducted for improper signs and units.

1. The reflection coefficient at a **short-circuited end** of a transmission line is..... [1]

2. On a Smith Chart, a movement **clockwise** corresponds to moving towards the [1]

3. The input impedance of a $\lambda/4$ (quarter-wave length) transmission line terminated with an open circuit is [1]

4. An unpolarized wave is incident from air into paraffin ($\epsilon_r = 3$). Find the incident angle for which the reflected wave is completely TE polarized. [2]

$$\Gamma = \frac{\eta_2 \cos(\theta_i) - \eta_1 \cos(\theta_t)}{\eta_2 \cos(\theta_i) + \eta_1 \cos(\theta_t)}, \quad \eta_1 = 377, \quad \eta_2 = \frac{377}{\sqrt{3}} = 217.66$$

5. A conductor is characterized by a conductivity, $\sigma = 100 \text{ S/m}$, and permittivity $\epsilon = 4 \epsilon_0$. At what angular frequency ω does the amplitude of the conduction current density equal the amplitude of the displacement current density? [2]

6. Write down Maxwell's equation, which is based on Faraday's law (Both integral and differential form). [2]

$$\nabla \times \mathbf{E} = 0 \quad \text{and} \quad \oint \mathbf{E} \cdot d\mathbf{s} = \frac{Q_c}{\epsilon}$$

$$\nabla \cdot \mathbf{E} = -\frac{\partial B}{\partial t}$$

$$\oint \mathbf{E} \cdot d\mathbf{l} = \mathbf{B}$$

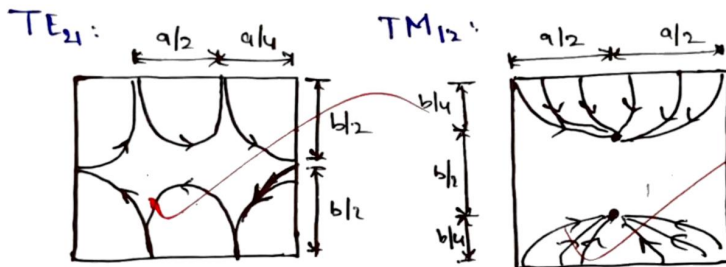
7. A rectangular waveguide filled with a dielectric of $\epsilon_r = 4$, with a cross-section of $5 \text{ cm} \times 2 \text{ cm}$, propagates the TM_{11} mode at 9 GHz . Find the cut-off wavelength [2]

$$f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \quad ; \quad c' = \frac{c}{\sqrt{\epsilon_r}} = \frac{c}{2}$$

$$\therefore \frac{9 \times 10^9}{4} \sqrt{\frac{1}{25} + \frac{1}{4}} \Rightarrow \boxed{f_c = 4.038 \times 10^8 \text{ Hz}}$$

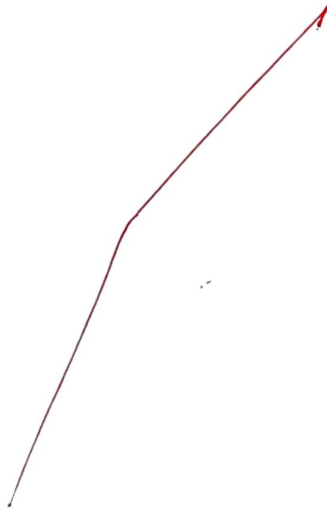
$\lambda_c = ?$

8. Draw the field pattern for TE_{21} and TM_{12} modes in a rectangular waveguide. [3]



9. Find the ratio of free space wavelength to wavelength in a conductor with $\sigma = 10^4 \text{ S/m}$ and $\mu_r = 15$ at 10 KHz . [3]

$$\frac{\lambda}{\lambda_g} =$$

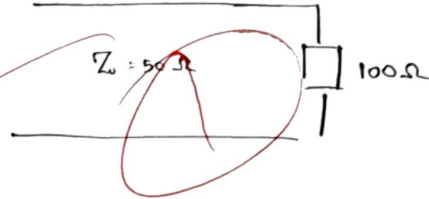


10. A lossless transmission line has a characteristic impedance of 50Ω and is terminated in a load of 100Ω . The magnitude of a voltage wave incident on the line is 20 V (rms) . Determine the maximum voltage and minimum current on the line. [4]

Min current occurs @ max voltage.

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{50}{150} = \frac{1}{3}$$

$$V_{SWR} = \frac{1 + \Gamma}{1 - \Gamma} = \frac{4/3}{2/3} = 2 = \frac{V_{max}}{V_{min}}$$



$20\text{ V rms} \rightarrow 20\sqrt{2}$ peak voltage.

$$\text{max voltage} = 20\sqrt{2}\text{ V}$$

$$\text{min current} = 0.5657\text{ A}$$

11. A uniform plane wave $\vec{E} = E_0 \hat{a}_y$ propagates in the x direction in a simple lossless medium characterized by $\epsilon_r = 9$, $\sigma = 0$ and $\mu_r = 4$. E_y is sinusoidal and has a frequency of 100 MHz . Its maximum value is 0.01 V/m at $t = 0$ and $x = 1/6$ meter. Obtain an instantaneous expression of H in terms of t and x . [4]

$$\vec{E} = E_0 \cos(\omega t - \beta x) \hat{a}_y ; H_0 = \frac{E_0}{\eta} ; \eta = \sqrt{\frac{\mu}{\epsilon}}$$

$$0.01 = E_0 \cos(200\pi \times 10^6 \times 0 - 1.2566 \times \frac{1}{6})$$

$$\omega = 2\pi f = 200\pi \times 10^6$$

$$\eta = 897.7 \times \sqrt{\frac{4}{9}}$$

$$\Rightarrow E_0 = 0.010223\text{ V/m}$$

$$\therefore H_0 = \frac{0.010223}{251.33} = 4.0677 \times 10^{-5}$$

$$\eta = 251.33$$

$$\beta = \frac{\omega}{v_p} = 1.2566$$

$$v_p = \frac{c}{\sqrt{\epsilon_r \mu_r}} = \frac{c}{6} = 5 \times 10^8$$

$$\vec{H} = (4.0677 \times 10^{-5}) \cos(200\pi \times 10^6 t - 1.2566 x) \hat{a}_z$$

7 ap

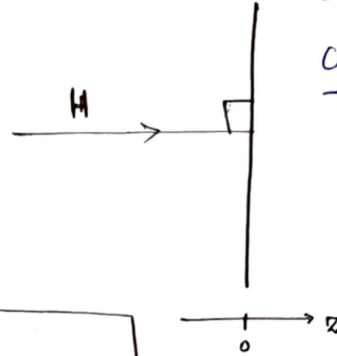
12. In free space $Z \leq 0$, a plane wave has $\mathbf{H} = 12 \cos(18 \times 10^7 t - \beta z) \hat{\mathbf{a}}_x$ (mA/m). The wave is incident normally on a lossless medium characterized by $\epsilon_r = 9$, and $\mu_r = 4$ in region $Z \geq 0$. Obtain reflected wave ($\mathbf{E}_r, \mathbf{H}_r$) and transmitted wave ($\mathbf{E}_t, \mathbf{H}_t$). [5]

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -0.2$$

$$\eta_1 = 377$$

$$\eta_2 = 251.33$$

$$\cos(90^\circ) = 1$$



20% of wave reflected.

80% of wave transmitted.

$$\begin{cases} \mathbf{H}_r = -2.4 \cos(18 \times 10^7 t - \beta z) \hat{\mathbf{a}}_x \text{ mA/m} \\ \mathbf{E}_r = (6.366 \times 10^{-3}) \cos(18 \times 10^7 t - \beta z) \hat{\mathbf{a}}_y \text{ mV/m} \end{cases}$$

2

$$\begin{cases} \mathbf{H}_t = 9.6 \cos(18 \times 10^7 t - \beta z) \hat{\mathbf{a}}_x \text{ mA/m} \\ \mathbf{E}_t = 0.0265 \cos(18 \times 10^7 t - \beta z) \hat{\mathbf{a}}_y \text{ mV/m} \end{cases}$$

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1. For rust at 2MHz, $\epsilon_r = 15$, $\sigma = 0.4 \times 10^{-2} \text{ S/m}$, and $\mu_r = 1$. Find the values of α , β , and phase velocity. [5]
2. A region $Y \geq 0$ consists of a dielectric medium. Region $Y < 0$ is a perfect conductor. For surface charge 4 nC/m^2 on the conductor and ϵ_r of the dielectric medium as 3, find E and D at the points
(A) Point (4,-2,1) (B) Point (-3,1,4) using boundary conditions. [5]
3. Antenna with impedance $40 + j30 \Omega$ is to be matched to a 100Ω lossless line with a shorted stub. Determine the following using the Smith chart: [10]
(a) The required stub admittance
(b) The distance between the stub and the antenna
(c) The stub length

Please mention both the solutions.
4. A uniform plane wave in air with $E = 8 \cos(\omega t - 4x - 3z) a_y \text{ V/m}$ is incident on a dielectric slab ($z \geq 0$) with $\epsilon_r = 2.5$, $\sigma = 0$, and $\mu_r = 1$. Find the following: [10]
(a) The polarization of the wave
(b) The angle of incidence
(c) The reflected E field
(d) The transmitted H field